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Write a C program to implement the LRU Page Replacement Algorithm. The program should accept a reference string and the number of frames. For each page request, identify whether it results in a Hit or a Fault. If a Fault occurs and the frames are full, replace the page that was accessed least recently. Display the frame status at each step and calculate the total number of Page Faults and Hits.

Input Format:

1. The first line of input contains an integer representing the number of pages in the reference string.
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format:

Select Language: C (GCC 9.2.0)

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int main() {
5     int n, frames_count, i, j, page_faults = 0, page_hits = 0;
6     int *ref_string, *frames, *last_used;
7     int is_fault, lru_index, min_last_used;
8
9     printf("Enter number of pages in reference string: \n");
10    scanf("%d", &n);
11
12    ref_string = (int *)malloc(n * sizeof(int));
13    printf("Enter the reference string: \n");
14    for (i = 0; i < n; i++) {
15        scanf("%d", &ref_string[i]);
16    }
17
18    printf("Enter number of frames: \n");
19    scanf("%d", &frames_count);
20
21    frames = (int *)malloc(frames_count * sizeof(int));
22    last_used = (int *)malloc(frames_count * sizeof(int));
23
24    for (i = 0; i < frames_count; i++) {
```

Implement the LRU Page Replacement Algorithm. The program should accept a reference string and the number of frames. For each page request, identify whether it results in a Hit or a Fault. If a Fault occurs and the frames are full, replace the page that was accessed least recently. Display the frame status at each step and calculate the total number of Page Faults and Hits.

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Output Format:

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```
19 frames[i] = -1;
20 last_used[i] = -1;
21 }
22
23 for (i = 0; i < n; i++) {
24     is_fault = 1;
25
26     for (j = 0; j < frames_count; j++) {
27         if (frames[j] == ref_string[i]) {
28             is_fault = 0;
29             page_hits++;
30             last_used[j] = i;
31             break;
32         }
33     }
34
35     if (is_fault) {
36         page_faults++;
37
38         int empty_frame = -1;
39         for (j = 0; j < frames_count; j++) {
40             if (frames[j] == -1) {
41                 empty_frame = j;
42                 break;
43             }
44         }
45     }
46 }
```

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Output Format:

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```
59 }
60
61 if (empty_frame != -1) {
62     frames[empty_frame] = ref_string[i];
63     last_used[empty_frame] = i;
64 } else {
65     lru_index = 0;
66     min_last_used = last_used[0];
67     for (j = 1; j < frames_count; j++) {
68         if (last_used[j] < min_last_used) {
69             min_last_used = last_used[j];
70             lru_index = j;
71         }
72     }
73     frames[lru_index] = ref_string[i];
74     last_used[lru_index] = i;
75 }
76
77 printf("Total Page Faults: %d\n", page_faults);
78 printf("Total Page Hits: %d\n", page_hits);
79
80 free(ref_string);
```

Custom Test

Success

Input :

5
1 2 3 4 5
5

Expected Output :

Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 5
Total Page Hits: 0

Output :

Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 5
Total Page Hits: 0